

[Coding Theory]

Problem-Solving Exercises

Coding Theory

Notations used throughout this test

- SNR: Signal-to-noise ratio.
- P_{eb} : Bit error probability.
- $H(X; Y)$: Mutual information of sources X and Y.
- P_{x_i, y_j} : Probability that symbol x_i is generated by source X and, at the same time, symbol y_j is generated by source Y.
- P_{x_i} : Probability that symbol x_i is generated by source X.
- P_{y_j} : Probability that symbol y_j is generated by source Y.
- $P_{y_j|x_i}$: Probability that symbol y_j is generated by source Y given the fact that symbol x_i has been generated by source X.

Question 1

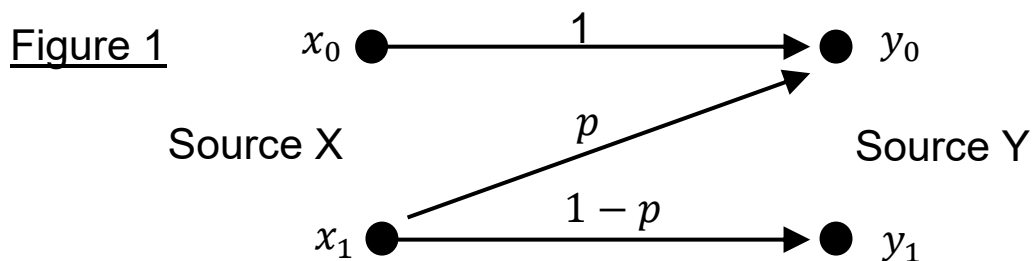
- (a) Explain why it is important, at the end of any convolutional encoding process, to terminate the trellis of the convolutional encoder in the zero state.

Also explain how communication engineers ensure that the trellis is terminated in the zero state.

You can use drawing(s) and/or equation(s) in your answer, if you wish to do so.

[10 marks]

- (b) We propose in this question to determine the capacity expression of the communication channel modelled in Figure 1. This channel will be hereafter referred to as *Z channel*.



Source X (channel input) generates a symbol $x_0 = 0$ with probability $P_{x_0} = \frac{1}{2}$ or a symbol $x_1 = 1$ with probability $P_{x_1} = \frac{1}{2}$.

Source Y (channel output) generates a symbol $y_0 = 0$ with probability P_{y_0} or a symbol $y_1 = 1$ with probability P_{y_1} .

The parameter p denotes the probability that symbol x_1 is transmitted erroneously. Note that symbol x_0 cannot be transmitted erroneously.

The channel capacity is given by the following generic expression:

$$C = H(X; Y) = \sum_{j=0}^1 \sum_{i=0}^1 P_{x_i, y_j} \cdot \log_2 \left(\frac{P_{y_j|x_i}}{P_{y_j}} \right).$$

(i) Find the expressions of probabilities $P_{y_0|x_1}$ and $P_{y_1|x_1}$, as a function of p .

Also, determine the values of probabilities $P_{y_0|x_0}$ and $P_{y_1|x_0}$.

[5 marks]

(ii) Find the expressions of both probabilities P_{y_0} and P_{y_1} as a function of p .

[5 marks]

(iii) Use the results obtained in (i) and (ii) to determine the expression of the capacity, C , of the Z channel as a function of p .

To answer, you need to remember that $u \cdot \log_2 u \rightarrow 0$ as $u \rightarrow 0$, i.e., we have $\lim_{u \rightarrow 0} (u \cdot \log_2 u) = 0$.

[5 marks]

(iv) Compute the values of the Z channel capacity obtained when $p = 0$ and $p = 1$. Briefly comment on your results.

[5 marks]

Question 2

- (a) Figure 2 shows several “ P_{eb} versus SNR” curves obtained over AWGN channel using various communication systems.

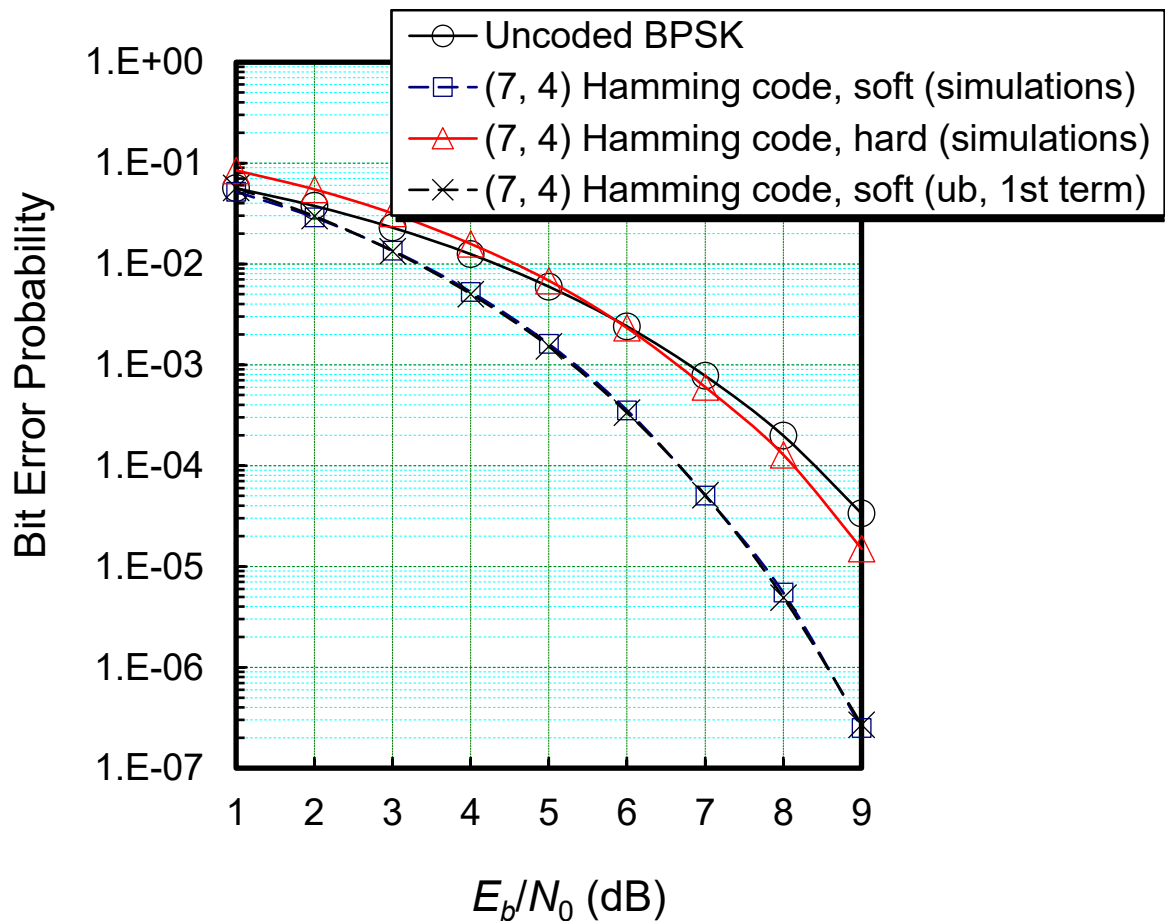


Figure 2

- (i) What is the equation used to graph the curve entitled “Uncoded BPSK”?

[4 marks]

- (ii) What is the equation used to graph the curve entitled “(7, 4) Hamming code, soft (ub, 1st term)”?

[4 marks]

(iii) Figure 2 indicates that the (7, 4) Hamming code performs better when the decoder uses soft decisions rather than hard decisions.

Does this result make sense?

Justify your answer.

[4 marks]

(iv) Are the results shown in Figure 2 independent of the algorithm used to decode the Hamming code?

Or, are they instead obtained under the implicit assumption that a particular decoding algorithm is employed?

Justify your answer.

[4 marks]

(v) Figure 2 shows that, over AWGN channel, the use of a (7, 4) Hamming code leads to an error performance improvement over an uncoded BPSK system.

What is the price paid for such performance improvement?

In other words, what sacrifice do we have to accept in order to improve the performance of a communication system using a (7, 4) Hamming code?

[4 marks]

- (b) We have seen in class that the use of an error-correcting code over AWGN channel provides an asymptotic coding gain G over uncoded BPSK which is given by $G = 10 \cdot \log_{10}(d_{min} \cdot R_c)$ dB.

This result explains why information theorists focused, for several decades, on the search for error-correcting codes that have the highest possible $d_{min} \cdot R_c$ product.

Did this simple approach lead to the discovery of codes that could perform near the capacity limit?

In other words, does the maximization of the $d_{min} \cdot R_c$ product guarantees near-capacity performance?

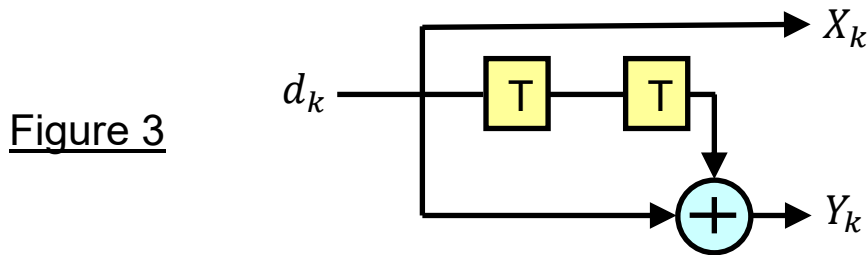
Justify your answer.

You can provide some example(s) to illustrate your answer and use drawing(s) and/or equation(s), if you wish to do so. However, remember that the length of your answer must not exceed half of an A4 page, including drawing(s) and equation(s).

[10 marks]

Question 3

Consider the convolutional encoder depicted in Figure 3. At each time $k \cdot T$, where T and k denote the encoder period and an integer, respectively, the encoder is fed with an information bit d_k and generates two coded bits X_k and Y_k .



- (a) Draw the trellis diagram for this encoder. **[10 marks]**
- (b) Use this trellis diagram to determine the free distance, d_{free} , for this encoder.

Show the path(s) in the trellis that correspond(s) to this free distance.

[6 marks]

- (c) Find the expression of the bit error probability obtained at high SNR using this convolutional code over a BPSK, AWGN channel.

What is the value of the asymptotic coding gain over uncoded BPSK?

[8 marks]

- (d) Do you think that the convolutional code studied in this question is optimal in terms of error performance?

Or, do you think that a better error performance can be achieved by employing another convolutional code with identical constraint length and coding rate?

Justify your answer.

[6 marks]

Question 4

- (a) We propose to design a channel code, hereafter referred to as *double-parity-check (DPC) code*, by combining two identical parity-check codes as explained below.

A message M is composed of $k = m^2$ information bits, where m is an arbitrary integer. Therefore, a message M can be viewed as a two-dimensional (2-D) array of size $m \times m$.

At the first step of encoding, each row of this $m \times m$ array is encoded into a $(m + 1)$ -bit codeword using a parity-check code. This row encoding thus generates a 2-D array composed of m rows and $(m + 1)$ columns.

At the second step of encoding, each column of this array is encoded into a $(m + 1)$ -bit codeword using, once again, a parity-check code. This column encoding finally produces a 2-D array of $(m + 1)$ rows and $(m + 1)$ columns that constitutes the codeword C associated with the $m \times m$ array M present at the encoder input.

Figure 4 illustrates the encoding procedure for a DPC code when $m = 3$.

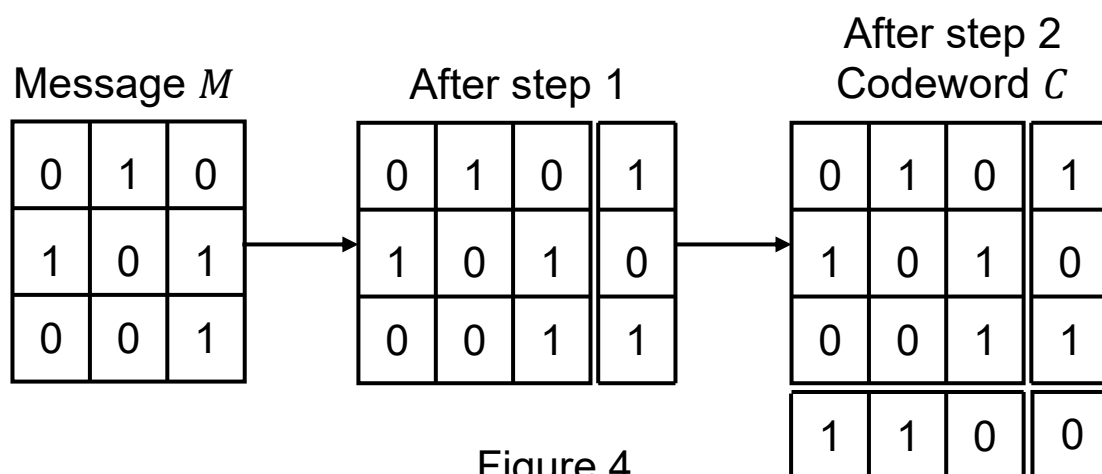


Figure 4

Throughout this question, assume that a DPC code is decoded using the maximum-likelihood (ML) algorithm.

(i) Is it realistic to assume ML decoding for a DPC code? Justify your answer.

[5 marks]

(ii) Determine the minimum distance d_{min} between codewords for a DPC code.

[10 marks]

(iii) Find the expression of the asymptotic coding gain G obtained using a DPC code over uncoded BPSK, as a function of the message length k .

[5 marks]

(b) In his landmark 1948 paper, Claude Shannon specified the characteristics that an error-correcting code must possess in order to display near-capacity error performance.

What are these characteristics?

Explain why modern codes, such as turbo and low-density parity-check (LDPC) codes, can indeed be considered as having these characteristics.

Explain why older codes, such as convolutional, Hamming, BCH, and Reed-Solomon codes, do not possess these characteristics.

[10 marks]

[END]